A phase strip and a Jacobsthal positive-defect suffix bound

27 August 2026

For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For a positive exponent word $v = (a_1, \dots, a_\ell)$, define

$$\mathrm{B}(v) := \sum_{i=1}^{\ell} a_i, \qquad \mathrm{C}(v) := \sum_{i=1}^{\ell} 3^{\ell-i} 2^{a_1+\cdots+a_{i-1}}. \tag{2}$$

We compare two exact paths of the same raw length and with the same endpoint. At a common boundary, write their state relation as

$$z = 2^{\Delta} x + c, \qquad u := \frac{c}{2^{\Delta}}, \tag{3}$$

where Δ is the difference between the masses of the two remaining suffixes. At every record-atom boundary, the established distortion theorem gives

$$\Delta \geq -1, \qquad -1 \leq u \leq 5. \tag{4}$$

The parity lattice gives even c for $\Delta = 0$, and odd c for $\Delta \geq 1$. At $\Delta = -1$, a nonempty remaining suffix has $c = -1/2$.

1 The phase strip

Every nontrivial record atom begins with exponent one:

$$B = 1\,|\,2, \qquad C = 1\,|\,113, \qquad D = 1\,|\,122. \tag{5}$$

The vertical bar marks the phase immediately after this first exponent.

Suppose that the right state after one of the tails

$$R \in \{2, 113, 122\} \tag{6}$$

is (d, c), and let v be the aligned alternative block. The state (Δ', c') before the tail satisfies

$$\Delta' = d + \mathrm{B}(v) - \mathrm{B}(R), \tag{7}$$

$$c' = \frac{2^{\Delta'}\,\mathrm{C}(R) + 2^{\mathrm{B}(v)}c - \mathrm{C}(v)}{3^{|R|}}. \tag{8}$$

Lemma 1 (lower strip at the first phase)**.** *At the phase marked in (5), every genuine distortion state satisfies*

$$\boxed{\frac{c'}{2^{\Delta'}} \geq -1.} \tag{9}$$

Proof. First take $R = 2$, and write $v = (a)$. If $d \geq 1$, the parity lattice sharpens (4) to

$$u \geq -1 + 2^{-d}. \tag{10}$$

Since $\Delta' = d + a - 2$, equations (7) and (8) give

$$u' = \frac{4}{3}u + \frac{1}{3} - \frac{2^{2-d-a}}{3} \geq -1 + \frac{2^{-d}(4 - 2^{2-a})}{3} \geq -1. \tag{11}$$

For $d = 0$, we have $u \geq 0$, and the same formula gives $u' \geq -1/3$. For $d = -1$, substitution of $c = -1/2$ gives

$$c' = -2^{a-3} - \frac{1}{3}, \tag{12}$$

which does not lie in the required dyadic parity lattice. Hence no such transition exists.

Now take $R = 113$ or $R = 122$. Their masses are five and their carries are respectively 19 and 23. Every positive three-letter word satisfies

$$\frac{\mathrm{C}(v)}{2^{\mathrm{B}(v)}} \leq \frac{19}{8}. \tag{13}$$

For $d \geq 2$, (10) and (8) therefore give

$$u' \geq \begin{cases} (-13 - 44 \cdot 2^{-d})/27, & R = 113, \\ (-9 - 44 \cdot 2^{-d})/27, & R = 122. \end{cases} \tag{14}$$

Both lower bounds are at least -1.

It remains to treat $d = -1, 0, 1$. The strip and parity lattice leave ten right states. Exact substitution gives the following minimum values of u'; a dash means that no transition exists:

d	c	$R = 113$	$R = 122$
-1	$-1/2$	$-7/8$	–
0	0	-1	0
0	2	2	–
0	4	–	21/4
1	-1	–	1/16
1	1	9/8	–
1	3	19/8	81/32
1	5	–	241/64
1	7	4	–
1	9	11/2	11/2.

(15)

For completeness, this table has a finite exact reduction. When $\Delta' < 3$, the total mass of v is fixed. When $\Delta' \geq 3$, write $v = (a, b_1, b_2)$. Reduce b_1, b_2 to positive residues modulo

$$\mathrm{ord}_{27}(2) = 18, \tag{16}$$

and, for each parity, reduce a to the first positive value with $\Delta' \geq 3$. These operations preserve transition admissibility and can only increase $\mathrm{C}(v)/2^{\mathrm{B}(v)}$, hence can only decrease u'. The resulting finite table is (15), and every entry is at least -1. □

2 The positive-defect suffix

Define the Jacobsthal number

$$J_\delta := \frac{2^\delta - (-1)^\delta}{3}. \tag{17}$$

Theorem 2 (Jacobsthal lower bound for a positive-defect suffix)**.** *Let $W = PU$, with $P \in \{B, C, D\}$, be an exact record-atom word from q to y. Let F be an exact word of the same raw length from r to y, and suppose*

$$\mathrm{B}(F) - \mathrm{B}(W) = \delta \geq 1. \tag{18}$$

Then

$$\boxed{r \geq 2^\delta q - J_\delta = 2^\delta q - \frac{2^\delta - (-1)^\delta}{3}.} \tag{19}$$

Proof. Write

$$r = 2^\delta q + c. \tag{20}$$

Both q and r are odd, so c is odd. Let a be the first exponent of F. Since the first exponent of W is one, the distortion state immediately after these two steps is

$$d = \delta + 1 - a, \qquad c_1 = \frac{3c + 1 - 2^\delta}{2^a}. \tag{21}$$

Lemma 1 gives $c_1/2^d \geq -1$. Therefore

$$3c + 1 - 2^\delta \geq -2^{a+d} = -2^{\delta+1}, \tag{22}$$

and hence

$$3c \geq -2^\delta - 1. \tag{23}$$

If δ is odd, (23) gives $c \geq -J_\delta$ directly. If δ is even, it gives

$$c \geq -J_\delta - \frac{2}{3}.$$

Since c and J_δ are odd integers, this again implies $c \geq -J_\delta$. Substitution in (20) proves (19). □

Corollary 3 (improved fixed-seed depth threshold)**.** *Let n be the least positive odd integer with an unbounded accelerated orbit. If a variable-depth competitor uses a suffix of positive defect δ, put*

$$r_\delta := 2^\delta q - J_\delta. \tag{24}$$

If its inverse prefix has length h, excess $t \geq T$, and start below n, then necessarily

$$\boxed{h > \log_{3/2}\left(\frac{r_\delta + 1}{2/3 + (n + 1/3)/2^T}\right).} \tag{25}$$

For $T = 3$, the denominator in (25) is $(3n + 17)/24$.

Proof. Theorem 2 supplies the suffix lower bound $r \geq r_\delta$. Apply the sharp all-excess prefix interval with r_δ in place of the suffix start and solve for h. □

The estimate (19) replaces the earlier bound $r \geq 2^\delta(q - 1)$. It removes approximately two thirds of the former uncertainty in the negative offset. The deeper positive-defect prefix branch is more constrained, but it is not yet excluded.